

# BRIAN A. FRENO

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## Education

**Texas A&M University**, College Station, TX

- |                                                                          |               |            |
|--------------------------------------------------------------------------|---------------|------------|
| • <b>Doctor of Philosophy</b> in Aerospace Engineering                   | December 2013 | GPA: 4.000 |
| • <b>Master of Science</b> (Thesis) in Aerospace Engineering             | May 2010      | GPA: 3.869 |
| • <b>Bachelor of Science</b> in Aerospace Engineering, Mathematics Minor | December 2008 | GPA: 3.425 |

## Work Experience

**Sandia National Laboratories**, Albuquerque, NM

**October 2015 – Present**

Principal Member of the Technical Staff, Intralevel 3

Verification, Validation, and Uncertainty Quantification

- Devised approaches to engineer features that, with machine-learning regression, can accurately predict the error incurred by reduced-order models and other approximate solutions to parameterized systems of nonlinear equations
- Formulated methods for computing symmetric triangle quadrature rules for arbitrary functions
- Developed novel code-verification techniques for computational fluid dynamics (CFD) with hypersonic reacting flow in thermochemical non-equilibrium, decomposing and non-decomposing ablation, and particle in cell with DSMC collisions
- Served as VVUQ reviewer for Oak Ridge National Laboratory and Los Alamos National Laboratory
- Lead the development of innovative code-verification and integration techniques for computational electromagnetics (CEM), which included the requirement-exceeding completion of an NNSA ASC Level 2 Milestone, as PI
- Strengthening the Academic Alliance with Texas A&M as adjunct professor through lecturing, mentoring, and recruiting
- Serving as reviewer and chair for NNSA Advanced Simulation and Computing programs and milestone

**Halliburton**, Houston, TX

**June 2014 – September 2015**

Senior Technical Professional

Production Enhancement – Advanced Computational Sciences

- Developed a parallel third-order-accurate compact incompressible viscous flow solver for non-uniform grids
- Invented an efficient mesh deformation algorithm for hydraulic fracture propagation that yielded a patent

**Texas A&M University**, College Station, TX

**Fall 2008 – Spring 2014**

Graduate Research Assistant

Department of Aerospace Engineering

- Developed reduced-order models for nonlinear structural dynamics and fluid mechanics for computational aeroelasticity and created software to produce 3D surface plots and movies  
Spring 2009 – Spring 2014
- Organized the Aerospace Engineering Study Abroad Program in Brazil, learned basic Portuguese  
Summer 2010
- Served as teaching assistant for junior-level propulsion class in Brazil  
Summer 2010
- Served as teaching assistant and occasional lecturer for graduate-level finite element course  
Spring 2009
- Served as grader for senior-level numerical simulation course  
Fall 2008

**NASA Marshall Space Flight Center**, Huntsville, AL

**Summers 2012 & 2013**

Graduate Student Researchers Program Fellow

Fluid Dynamics Branch

- Conducted CFD reduced-order modeling research as part of NASA Graduate Student Researchers Program Fellowship

**Lockheed Martin Missiles and Fire Control**, Orlando, FL

**Summers 2007 & 2008**

Summer Intern

Aerodynamics Department

- Developed 2D and 3D, steady and unsteady, rigid and flexible panel codes and created GUI
- Produced and analyzed aerodynamic performance plots of missile CFD, DATCOM, and wind tunnel data

**Standard Aero**, San Antonio, TX

**Summers 2005 & 2006**

Summer Intern

Reliability Engineering

- Developed algorithms, implementations, and communication strategies for Reliability Centered Maintenance

## Journal Articles (Primary Author)

- B. Freno, W. McDoniel, C. Moore, N. Matula, “Code-Verification Techniques for Particle-in-Cell Simulations with Direct Simulation Monte Carlo Collisions,” *under review*, [arXiv:2602.15130](#)
- B. Freno, N. Matula, J. Bishop, “A Note on the Convergence of Symmetric Triangle Quadrature Rules,” *Journal of Computational Physics* 548 (2026), [10.1016/j.jcp.2025.114564](#)
- B. Freno, N. Matula, R. Pfeiffer, V. Dang, “Code-verification techniques for an arbitrary-depth electromagnetic slot model,” *Engineering Analysis with Boundary Elements* 178 (2025), [10.1016/j.enganabound.2025.106275](#)
- B. Freno, N. Matula, R. Pfeiffer, E. Dohme, J. Kotulski, “Manufactured solutions for an electromagnetic slot model,” *Journal of Computational Physics* 516 (2024), [10.1016/j.jcp.2024.113343](#)
- B. Freno, N. Matula, “Code-verification techniques for the method-of-moments implementation of the combined-field integral equation,” *Journal of Computational Physics* 488 (2023), [10.1016/j.jcp.2023.112231](#)
- B. Freno, N. Matula, “Code-verification techniques for the method-of-moments implementation of the magnetic-field integral equation,” *Journal of Computational Physics* 478 (2023), [10.1016/j.jcp.2023.111959](#)
- B. Freno, N. Matula, “Code verification for practically singular equations,” *Journal of Computational Physics* 470 (2022), [10.1016/j.jcp.2022.111581](#)
- B. Freno, B. Carnes, V. Brunini, N. Matula, “Nonintrusive manufactured solutions for non-decomposing ablation in two dimensions,” *Journal of Computational Physics* 463 (2022), [10.1016/j.jcp.2022.111237](#)
- B. Freno, N. Matula, J. Owen, W. Johnson, “Code-verification techniques for the method-of-moments implementation of the electric-field integral equation,” *Journal of Computational Physics* 451 (2022), [10.1016/j.jcp.2021.110891](#)
- B. Freno, N. Matula, W. Johnson, “Manufactured solutions for the method-of-moments implementation of the electric-field integral equation,” *Journal of Computational Physics* 443 (2021), [10.1016/j.jcp.2021.110538](#)
- B. Freno, W. Johnson, B. Zinser, D. Wilton, F. Vipiana, S. Campione, “Characterization and integration of the singular test integrals in the method-of-moments implementation of the electric-field integral equation,” *Engineering Analysis with Boundary Elements* 124 (2021), [10.1016/j.enganabound.2020.12.015](#)
- B. Freno, B. Carnes, N. Matula, “Nonintrusive manufactured solutions for ablation,” *Physics of Fluids* 33 (2021), [10.1063/5.0037245](#)
- B. Freno, B. Carnes, V. Weirs, “Code-verification techniques for hypersonic reacting flows in thermochemical nonequilibrium,” *Journal of Computational Physics* 425 (2021), [10.1016/j.jcp.2020.109752](#)
- B. Freno, W. Johnson, B. Zinser, S. Campione, “Symmetric triangle quadrature rules for arbitrary functions,” *Computers & Mathematics with Applications* 79, no. 10 (2020), [10.1016/j.camwa.2019.12.021](#)
- B. Freno, K. Carlberg, “Machine-learning error models for approximate solutions to parameterized systems of nonlinear equations,” *Computer Methods in Applied Mechanics and Engineering* 348 (2019), [10.1016/j.cma.2019.01.024](#)
- B. Freno, N. Matula, R. Fontenot, P. Cizmas, “The use of dynamic basis functions in proper orthogonal decomposition,” *Journal of Fluids and Structures* 54 (2015), [10.1016/j.jfluidstructs.2014.11.009](#)
- B. Freno, P. Cizmas, “A proper orthogonal decomposition method for nonlinear flows with deforming meshes,” *International Journal of Heat and Fluid Flow* 50 (2014), [10.1016/j.ijheatfluidflow.2014.07.001](#)
- B. Freno, T. Brenner, P. Cizmas, “Using proper orthogonal decomposition to model off-reference flow conditions,” *International Journal of Non-Linear Mechanics* 54 (2013), [10.1016/j.ijnonlinmec.2013.03.007](#)
- B. Freno, P. Cizmas, “An investigation into the significance of the non-linear terms in the equations of motion for a cantilevered beam,” *International Journal of Non-Linear Mechanics* 47, no. 3 (2012), [10.1016/j.ijnonlinmec.2012.01.002](#)
- B. Freno, P. Cizmas, “A computationally efficient non-linear beam model,” *International Journal of Non-Linear Mechanics* 46, no. 6 (2011), [10.1016/j.ijnonlinmec.2011.03.010](#)

## Patent

- B. Freno, S. Madasu, A. Lin, Simulating hydraulic fracture propagation using dynamic mesh deformation, US Patent No. 10,947,820, Issued March 16, 2021

## Honors & Awards

- **American Institute of Aeronautics and Astronautics** Associate Fellow Jan. 2024
- **American Society of Mechanical Engineers**
  - Heat Transfer Division Outstanding Reviewer Nov. 2021
  - Reviewer of the Year, *Journal of Verification, Validation, and Uncertainty Quantification* Dec. 2020
- **Halliburton** Invention Disclosure Award Sep. 2015
- **NASA** Graduate Student Researchers Program Fellowship Sep. 2011
- **Sandia National Laboratories**
  - Individual Performance Award Aug. 2017 Dec. 2018 Sep. 2020 Jun. 2023 May 2024 Oct. 2024 May 2026
  - Thunderbird Kudos Award Oct. 2022 Nov. 2022 Jun. 2025 Nov. 2025 Feb. 2026 Mar. 2026
  - Employee Recognition Awards Nominee Jan. 2020 Feb. 2021 Feb. 2023 Feb. 2024 Feb. 2025 Feb. 2026
  - High Performance Incentive Plan Award (ended 2023) Dec. 2019 Oct. 2020 Oct. 2021 Oct. 2022 Oct. 2023
  - Innovation and Intellectual Property Award Nov. 2021
  - Critical Skills Retention Incentive Sep. 2021
- **Texas A&M University Department of Aerospace Engineering**
  - Outstanding Young Aerospace Engineer Distinguished Alumni Award Nov. 2022
  - Outstanding Achievement Award – Aerodynamics & Propulsion May 2014
  - Outstanding Doctoral Student Award May 2014
  - Boeing Graduate Fellowship Jan. 2009
  - Stan H. Lowy Award for Excellence in Aerospace Design Dec. 2008

## Professional Societies

- Associate Fellow, **American Institute of Aeronautics and Astronautics**
- Member, **American Society of Mechanical Engineers**
- Member, **Society for Industrial and Applied Mathematics**

## Theses

- B. Freno, Reduced-order models for computational aeroelasticity, PhD dissertation, Texas A&M University, Dec. 2013
- B. Freno, An efficient nonlinear structural dynamics solver for use in computational aeroelastic analysis, Master's thesis, Texas A&M University, May 2010

## Presented Conference Papers

- B. Freno, B. Carnes, N. Matula, "Nonintrusive manufactured solutions for ablation," *2021 AIAA SciTech Forum*, AIAA Paper 2021-1174, Jan. 2021
- B. Freno, B. Carnes, V. Weirs, "Code-verification techniques for hypersonic reacting flows in thermochemical nonequilibrium," *2019 AIAA Aviation Forum*, AIAA Paper 2019-3705, Dallas, TX, Jun. 2019
- B. Freno, N. Matula, R. Fontenot, P. Cizmas, "The use of dynamic basis functions in proper orthogonal decomposition," *2014 AIAA SciTech Forum*, AIAA Paper 2014-1436, National Harbor, MD, Jan. 2014
- B. Freno, P. Cizmas, "A proper orthogonal decomposition method for nonlinear flows with deforming meshes," *51<sup>st</sup> AIAA Aerospace Sciences Meeting*, AIAA Paper 2013-0055, Grapevine, TX, Jan. 2013
- B. Freno, T. Brenner, P. Cizmas, "Proper orthogonal decomposition applied to the Reynolds-averaged Navier-Stokes equations," *50<sup>th</sup> AIAA Aerospace Sciences Meeting*, AIAA Paper 2012-314, Nashville, TN, Jan. 2012
- B. Freno, R. Brown, P. Cizmas, "The role of structural nonlinearities in wind turbine blade aeroelastic analysis," *49<sup>th</sup> AIAA Aerospace Sciences Meeting*, AIAA Paper 2011-995, Orlando, FL, Jan. 2011

## Professional Service

- **Deputy Director:** AIAA Aerospace Sciences Group Sep. 2025 – Present
- **Adjunct Professor:** Texas A&M University Department of Aerospace Engineering Dec. 2022 – Present
- **Associate Editor:** ASME Journal of Verification, Validation, and Uncertainty Quantification Feb. 2021 – Present
- **Journal Reviewer:** Nov. 2011 – Present
  - AIAA Journal
  - AIAA Journal of Spacecraft and Rockets
  - ASME Journal of VVUQ (5)
  - Computational and Applied Mathematics (2)
  - Computer Methods in Applied Mechanics and Engineering
  - Engineering Analysis with Boundary Elements (2)
  - IEEE Transactions on Antennas & Propagation
  - Int. Journal for Numerical Methods in Fluids
  - Inverse Problems in Science & Engineering
  - Journal of Computational Physics (2)
  - Physics of Fluids (3)
- **Minisymposium/Session Organizer:**
  - AIAA SciTech: Verification techniques in computational physics Jan. 2026
  - WCCM/PANACM: Verification techniques in computational physics and applied mathematics Jul. 2024
  - ASME VVUQ Symposium: Computational electromagnetics, plasma, radiation May 2024
  - WCCM/APCOM: Verification techniques in computational physics and applied mathematics Jul. 2022
  - ASME VVUQ Symposium: VVUQ for advanced manufacturing, plasma, radiation transport May 2022
  - SIAM UQ: Verification techniques in computational physics and applied mathematics Apr. 2022
  - ASME V&V Symposium: VVUQ for artificial intelligence and machine learning models May 2021
  - SIAM CSE: Numerical methods for integral and integro-differential equations Mar. 2021
  - WCCM/ECCOMAS: Verification techniques in computational mechanics and applied mathematics Jan. 2021
  - WCCM/ECCOMAS: Improving predictive capabilities through model error quantification Jan. 2021
  - ASME V&V Symposium: VVUQ for computational electromagnetics, plasma, radiation transport May 2020
- **Committees:**
  - ASME VVUQ in Computational Fluid Dynamics and Heat Transfer Subcommittee Jun. 2025 – Present
  - University of New Mexico Hospital Patient and Family Advisory Committee Jun. 2024 – Present
  - AIAA Fluid Dynamics Technical Committee, CFD Subcommittee May 2022 – May 2025
    - Organizer of AIAA Aviation Forum Flow Visualization Showcase Jun. 2023
    - Associate organizer for AIAA Aviation (CFD: Reduced-order modeling & CFD: VVUQ) Jun. 2023
    - Session chair for AIAA SciTech and Aviation Jun. 2022 – Jan. 2024
    - Reviewer for AIAA Aviation 2024 / SciTech 2025 Best Paper Award Mar. 2025
    - Associate organizer for AIAA SciTech Jan. 2026
    - Reviewer for AIAA SciTech and Aviation Jan. 2023 – Jan. 2026
- **Program Reviewer:**
  - NNSA ASC PSAAP IV Tri-Lab Sponsor Team Aug. 2025 – Present
  - Sandia National Laboratories Laboratory Directed Research & Development (LDRD) May 2018 – May 2026
    - Nuclear Deterrence Investment Area
    - Computing and Information Sciences Investment Area
    - Exploratory Express Investment Area
  - AIAA Region IV Student Conference Mar. 2026
  - Review chair, NNSA ASC Level 2 Milestone: Multi-Fidelity & ROM Methods for Reentry Oct. 2024 – Sep. 2025
  - NNSA ASC Predictive Science Academic Alliance Program (PSAAP) III Review Team Jun. 2020 – Jun. 2025
  - Los Alamos National Laboratory ASC Level 2 Milestone Aug. 2024
  - Texas A&M Engineering Project Showcase Apr. 2022
- **Mentor:** Early-career individuals and students, with an emphasis on under-represented groups in STEM

## Presentations

|                                                                                                          |           |
|----------------------------------------------------------------------------------------------------------|-----------|
| • World Congress on Computational Mechanics / ECCOMAS, Munich, Germany                                   | Jul. 2026 |
| • ASME Verification, Validation, and Uncertainty Quantification Symposium, Nashville, TN                 | May 2026  |
| • AIAA SciTech Forum, Orlando, FL                                                                        | Jan. 2026 |
| • Direct Simulation Monte Carlo Workshop, Santa Fe, NM                                                   | Sep. 2025 |
| • General Atomics Aeronautical Systems                                                                   | Sep. 2025 |
| • IEEE Int. Sym. on Antennas and Propagation and North American Radio Sci. Meeting, Ottawa, ON           | Jul. 2025 |
| • ASME Verification, Validation, and Uncertainty Quantification Symposium, College Station, TX           | Apr. 2025 |
| • SIAM Conference on Computational Science and Engineering, Fort Worth, TX                               | Mar. 2025 |
| • Texas A&M University Aerospace Practitioner and Professional Engr. Lecture Series, College Station, TX | Sep. 2024 |
| • World Congress on Comp. Mechanics / Pan American Congress on Comp. Mechanics, Vancouver, BC            | Jul. 2024 |
| • ASME Verification, Validation, and Uncertainty Quantification Symposium, College Station, TX           | May 2024  |
| • ASME International Mechanical Engineering Congress and Exposition, New Orleans, LA                     | Nov. 2023 |
| • IEEE Int. Sym. on Antennas and Propagation and North American Radio Sci. Meeting, Portland, OR         | Jul. 2023 |
| • ASME Verification, Validation, and Uncertainty Quantification Symposium, Baltimore, MD                 | May 2023  |
| • World Congress on Computational Mechanics / Asian Pacific Congress on Computational Mechanics          | Jul. 2022 |
| • IEEE Int. Sym. on Antennas and Propagation and North American Radio Sci. Meeting, Denver, CO           | Jul. 2022 |
| • Sandia National Laboratories Engineering Sciences Summer Institute Seminar Series                      | Jun. 2022 |
| • ASME Verification, Validation, and Uncertainty Quantification Symposium, College Station, TX           | May 2022  |
| • SIAM Conference on Uncertainty Quantification, Atlanta, GA                                             | Apr. 2022 |
| • Texas A&M University Industrial and Applied Mathematics Seminar Series, College Station, TX            | Jan. 2022 |
| • Texas A&M University Aerospace Practitioner and Professional Engr. Lecture Series, College Station, TX | Jan. 2022 |
| • International Conference on Electromagnetics in Advanced Applications                                  | Aug. 2021 |
| • Sandia National Laboratories Engineering Sciences Summer Institute Seminar Series                      | Jul. 2021 |
| • Oak Ridge National Laboratory Computational Sciences and Engineering Division                          | Jun. 2021 |
| • ASME Verification and Validation Symposium: VVUQ for Fluid Dynamics, Heat Transfer, Electromagnetics   | May 2021  |
| • Texas A&M University Aerospace Engineering Seminar Series                                              | Apr. 2021 |
| • SIAM Conference on Computational Science and Engineering                                               | Mar. 2021 |
| • World Congress on Computational Mechanics / European Congress on Computational Methods                 | Jan. 2021 |
| • AIAA SciTech Forum                                                                                     | Jan. 2021 |
| • IEEE International Symposium on Antennas and Propagation and North American Radio Science Meeting      | Jul. 2020 |
| • Texas A&M University Aerospace Engineering Seminar Series, College Station, TX                         | Jan. 2020 |
| • Texas A&M University College of Engineering, College Station, TX                                       | Sep. 2019 |
| • Sandia National Laboratories Machine Learning and Deep Learning Workshop, Albuquerque, NM              | Aug. 2019 |
| • AIAA Aviation Forum, Dallas, TX                                                                        | Jun. 2019 |
| • Sandia National Laboratories Engineering Sciences External Review Board, Albuquerque, NM               | Apr. 2019 |
| • Texas A&M University College of Engineering, College Station, TX                                       | Mar. 2019 |
| • Texas A&M University Industrial and Applied Mathematics Seminar Series, College Station, TX            | Mar. 2019 |
| • SIAM Conference on Computational Science and Engineering, Spokane, WA                                  | Feb. 2019 |
| • Sandia National Laboratories Center for Computing Research Seminar, Albuquerque, NM                    | Oct. 2018 |
| • World Congress on Computational Mechanics, New York, NY                                                | Jul. 2018 |
| • SIAM Conference on Uncertainty Quantification, Anaheim, CA                                             | Apr. 2018 |
| • University of Florida and Eglin AFB Research and Engineering Education Facility Campus, Shalimar, FL   | May 2014  |
| • AIAA SciTech Forum, National Harbor, MD                                                                | Jan. 2014 |
| • AIAA Aerospace Sciences Meeting, Grapevine, TX                                                         | Jan. 2013 |
| • AIAA Aerospace Sciences Meeting, Nashville, TN                                                         | Jan. 2012 |
| • AIAA Aerospace Sciences Meeting, Orlando, FL                                                           | Jan. 2011 |
| • University of Campinas (Unicamp), Campinas, São Paulo, Brazil                                          | Jul. 2010 |



## Teaching (Guest Lecturer)

- TAMU ENGR 681-602: Professional Development for Non-Academic Career Path Doctoral Students Fall 2019
- TAMU AERO 306: Aerospace Structural Analysis II Fall 2013
- TAMU AERO 430: Numerical Simulation Fall 2013
- TAMU MEMA 646: Introduction to the Finite Element Method Spring 2012

## Additional Publications (Secondary Author)

- J. Ray, S. Kieweg, D. Dinzl, B. Carnes, V. Weirs, **B. Freno**, M. Howard, T. Smith, I. Nompelis, G. Candler, Estimation of inflow uncertainties in laminar hypersonic double-cone experiments, *AIAA Journal* 58 (2020), doi:10.2514/1.J059033
- S. Reddy, **B. Freno**, P. Cizmas, S. Gokaltun, D. McDaniel, G. Dulikravich, Constrained reduced-order models based on proper orthogonal decomposition, *Computer Methods in Applied Mechanics and Engineering* 321 (2017), doi:10.1016/j.cma.2017.03.038
- A. Krueger, B. Lance, **B. Freno**, R. Wagnild, Verification Studies of the Multi-Fidelity Toolkit, *2022 AIAA SciTech Forum*, AIAA Paper 2022-2009, San Diego, CA, Jan. 2022
- B. Lance, A. Krueger, **B. Freno**, R. Wagnild, Validation Study of the Multi-Fidelity Toolkit, *2022 AIAA SciTech Forum*, AIAA Paper 2022-1574, San Diego, CA, Jan. 2022
- J. Ray, S. Kieweg, D. Dinzl, B. Carnes, V. Weirs, **B. Freno**, M. Howard, T. Smith, I. Nompelis, G. Candler, Estimation of inflow uncertainties in laminar hypersonic double-cone experiments, *2019 AIAA SciTech Forum*, AIAA Paper 2019-2279, San Diego, CA, Jan. 2019
- S. Kieweg, J. Ray, V. Weirs, B. Carnes, D. Dinzl, **B. Freno**, M. Howard, E. Phipps, W. Rider, T. Smith, Validation assessment of hypersonic double-cone flow simulations using uncertainty quantification, sensitivity analysis, and validation metrics, *2019 AIAA SciTech Forum*, AIAA Paper 2019-2278, San Diego, CA, Jan. 2019
- F. Carpenter, T. Brenner, **B. Freno**, P. Cizmas, A reduced-order model for turbomachinery flows using proper orthogonal decomposition, *ASME Turbo Expo 2013*, GT2013-94914, San Antonio, TX, Jun. 2013
- P. Cizmas, **B. Freno**, T. Brenner, G. Worley, A high-fidelity nonlinear aeroelastic model for aircraft with large wing deformations, *International Forum on Aeroelasticity and Structural Dynamics*, IFASD-2009-098, Seattle, WA, Jun. 2009

## Research Experience

- **Physics Disciplines**
  - Computational fluid dynamics (CFD)
    - Compressible and incompressible
    - Viscous and inviscid
  - Ablation and heat transfer
  - Nonlinear structural dynamics
  - Aeroelasticity
  - Computational electromagnetics (CEM)
- **Meshing**
  - Elliptic and Schwarz–Christoffel grid generation
  - Mesh deformation
- **Surrogate Modeling**
  - Reduced-order modeling
  - Proper orthogonal decomposition
  - Machine learning
- **Numerical Methods**
  - Code verification
  - Post-processing
  - Numerical integration

## Student Activities and Service

- Sigma Gamma Tau (National Aerospace Engineering Honor Society) – President, Vice President (Texas A&M Chapter)
- American Institute of Aeronautics and Astronautics – Chair, Vice Chair (Texas A&M Chapter)
- Texas A&M University Student Engineers' Council – Legislation Committee
- Texas A&M University Student Senate – Caucus Leader and Senator for College of Engineering
- Texas A&M University Wind Symphony – Performed in Carnegie Hall and Europe